

Module A: CG for non-quadratic f (“nonlinear CG”)

We now consider *general* smooth unconstrained minimization

$$\min_{x \in \mathbb{R}^n} f(x), \quad g_k := \nabla f(x_k).$$

Nonlinear CG keeps the same **two-line skeleton** as quadratic CG:

$$x_{k+1} = x_k + \alpha_k d_k, \quad d_{k+1} = -g_{k+1} + \beta_k d_k,$$

but β_k must be chosen *without* knowing a constant Hessian Q .

The “ideal” quadratic formula. For $f(x) = \frac{1}{2}x^\top Qx - b^\top x$ with $Q \succ 0$ and exact line search,

$$\beta_k^{\text{quad}} = \frac{g_{k+1}^\top Q d_k}{d_k^\top Q d_k}.$$

In general Q is replaced by **curvature information along the step**: let $y_k := g_{k+1} - g_k$.

Hestenes–Stiefel (HS), Polak–Ribière (PR), and Fletcher–Reeves (FR) Starting with the secant-like proxy $Qd_k \approx y_k/\alpha_k$, consider the ideal quadratic case:

$$\beta_k^{\text{HS}} := \frac{g_{k+1}^\top y_k}{d_k^\top y_k} = \frac{g_{k+1}^\top y_k}{\underbrace{d_k^\top g_{k+1} - d_k^\top g_k}_{=0}} = \frac{g_{k+1}^\top y_k}{g_k^\top g_k} = \beta_k^{\text{PRP}} := \frac{g_{k+1}^\top y_k}{g_k^\top g_k} = \frac{g_{k+1}^\top g_{k+1} - \overbrace{g_{k+1}^\top g_k}^{=0}}{g_k^\top g_k} = \beta_k^{\text{FR}} := \frac{g_{k+1}^\top g_{k+1}}{g_k^\top g_k}$$

Practice: Textbook 10.6 (Krylov subspaces & CG optimality)

Consider the algorithm

$$x^{(k+1)} = x^{(k)} + \alpha_k d^{(k)}, \quad x^{(0)} = 0,$$

applied to the quadratic

$$f(x) = \frac{1}{2}x^\top Qx - b^\top x, \quad Q \succ 0, \quad g^{(k)} = \nabla f(x^{(k)}) = Qx^{(k)} - b,$$

with search directions generated by

$$d^{(k+1)} = a_k g^{(k+1)} + b_k d^{(k)}, \quad d^{(-1)} = 0,$$

where $a_k, b_k \in \mathbb{R}$ are constants.

(a) Define $\mathcal{V}_k := \text{span}\{b, Qb, \dots, Q^{k-1}b\}$ (Krylov subspace of order k). Show that $d^{(k)} \in \mathcal{V}_{k+1}$ and $x^{(k)} \in \mathcal{V}_k$.

(b) In light of (a), what does CG “optimize” with respect to $\mathcal{V}_k$?

Answer: (a) **Induction.** Base: $\mathcal{V}_0 = \{0\}$, $\mathcal{V}_1 = \text{span}\{b\}$. Since $x^{(0)} = 0 \in \mathcal{V}_0$ and $g^{(0)} = -b \in \mathcal{V}_1$, and $d^{(0)} = a_{-1}g^{(0)} + b_{-1}d^{(-1)}$ (or simply take $d^{(0)}$ as a linear comb. of $g^{(0)}$ and $d^{(-1)} = 0$), we get $d^{(0)} \in \mathcal{V}_1$.

Inductive step: assume $x^{(k)} \in \mathcal{V}_k$ and $d^{(k)} \in \mathcal{V}_{k+1}$. Then $Qx^{(k)} \in Q\mathcal{V}_k \subseteq \mathcal{V}_{k+1}$, hence $g^{(k)} = Qx^{(k)} - b \in \mathcal{V}_{k+1}$. Update gives $x^{(k+1)} = x^{(k)} + \alpha_k d^{(k)} \in \mathcal{V}_k + \mathcal{V}_{k+1} = \mathcal{V}_{k+1}$. Finally $g^{(k+1)} = Qx^{(k+1)} - b \in \mathcal{V}_{k+2}$, so $d^{(k+1)} = a_k g^{(k+1)} + b_k d^{(k)} \in \mathcal{V}_{k+2} + \mathcal{V}_{k+1} = \mathcal{V}_{k+2}$.

(b) **Krylov-subspace optimality.** In (exact-arithmetic) CG with exact line search and Q -conjugate directions,

$$x^{(k)} \in \mathcal{V}_k \quad \text{and} \quad x^{(k)} = \arg \min_{x \in \mathcal{V}_k} f(x).$$

Equivalently, the residual/gradient is orthogonal to the whole Krylov subspace: $g^{(k)} \perp \mathcal{V}_k$ (Euclidean inner product), so $x^{(k)}$ is the best approximation to $x^* = Q^{-1}b$ over $\mathcal{V}_k$ (also interpretable as a best approximation in the Q -energy norm).

Module B: Quasi-Newton methods (Ch. 11.2–11.5)

Goal: approximate Newton without explicitly forming $\nabla^2 f(x)$.

Quasi-Newton form. At iterate $x^{(k)}$, let $g^{(k)} = \nabla f(x^{(k)})$ and choose

$$d^{(k)} = -H_k g^{(k)}, \quad \alpha_k = \arg \min_{\alpha \geq 0} f(x^{(k)} + \alpha d^{(k)}), \quad x^{(k+1)} = x^{(k)} + \alpha_k d^{(k)} \Leftrightarrow \Delta x^{(k)} = \alpha_k d^{(k)}.$$

Desired Properties: $H_k = H_k^\top$ and $H_k \succ 0$ (so $d^{(k)\top} g^{(k)} < 0$).

Secant (curvature-matching) condition. Define $\Delta x^{(k)} := x^{(k+1)} - x^{(k)}$, $\Delta g^{(k)} := g^{(k+1)} - g^{(k)}$.

Quasi-Newton updates enforce the *secant equation*

$$H_{k+1} \Delta g^{(k)} = \Delta x^{(k)} \quad (\text{equivalently } B_{k+1} \Delta x^{(k)} = \Delta g^{(k)} \text{ if } B_k \approx \nabla^2 f).$$

In the quadratic + exact line search case, satisfying these secant equations yields a conjugate-direction behavior (and solves in $\leq n$ steps).

Classic updates (what they are + relationship).

- **Rank-one correction (Single-Rank Symmetric):** symmetric rank-one change to satisfy $H_{k+1} \Delta g^{(k)} = \Delta x^{(k)}$,

$$H_{k+1} = H_k + \frac{(\Delta x^{(k)} - H_k \Delta g^{(k)})(\Delta x^{(k)} - H_k \Delta g^{(k)})^\top}{\Delta g^{(k)\top}(\Delta x^{(k)} - H_k \Delta g^{(k)})}.$$

Fast/cheap, but may lose $H_{k+1} \succ 0$. (When we can guarantee $H_{k+1} \succ 0$ for SRS? [Textbook 11.7])

- **DFP (rank-two update for H):**

$$H_{k+1}^{\text{DFP}} = H_k + \frac{\Delta x^{(k)} \Delta x^{(k)\top}}{\Delta x^{(k)\top} \Delta g^{(k)}} - \frac{H_k \Delta g^{(k)} \Delta g^{(k)\top} H_k}{\Delta g^{(k)\top} H_k \Delta g^{(k)}}.$$

Preserves SPD under the curvature condition $\Delta x^{(k)\top} \Delta g^{(k)} > 0$.

- **BFGS (rank-two update for B ; complementary/dual to DFP):**

$$B_{k+1}^{\text{BFGS}} = B_k - \frac{B_k \Delta x^{(k)} \Delta x^{(k)\top} B_k}{\Delta x^{(k)\top} B_k \Delta x^{(k)}} + \frac{\Delta g^{(k)} \Delta g^{(k)\top}}{\Delta g^{(k)\top} \Delta x^{(k)}}.$$

BFGS swap roles of $\Delta x^{(k)}$ and $\Delta g^{(k)}$ in DFP. **More robust for sloppy line search and thus more efficient than DFP.**

Practice: Textbook 11.6 (scaled steepest descent: when is $\alpha_k > 0$?)

Consider

$$x^{(k+1)} = x^{(k)} - \alpha_k M_k \nabla f(x^{(k)}), \quad M_k = \begin{bmatrix} 1 & 0 \\ 0 & a \end{bmatrix}, \quad a \in \mathbb{R},$$

with exact line search

$$\alpha_k = \arg \min_{\alpha \geq 0} f(x^{(k)} - \alpha M_k \nabla f(x^{(k)})).$$

Suppose at some iteration k we have $\nabla f(x^{(k)}) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Find the *largest range of values of a* that guarantees $\alpha_k > 0$

for any C^1 function f .

Answer: Let $g := \nabla f(x^{(k)})$ and the search direction be $d := -M_k g$. A sufficient (and here necessary, for a uniform guarantee over all C^1 functions) condition for $\alpha_k > 0$ is that d is a **descent direction**:

$$g^\top d < 0.$$

Here $g = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $M_k = \text{diag}(1, a)$, so

$$g^\top d = -g^\top M_k g = -\begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & a \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = -(1+a).$$

Thus $g^\top d < 0 \iff 1+a > 0 \iff a > -1$. Therefore the largest range guaranteeing $\alpha_k > 0$ for any f is

$$\boxed{a \in (-1, \infty)}.$$

(If $a = -1$, then $g^\top d = 0$ and $\alpha_k = 0$ can occur; if $a < -1$, the direction is ascent.)